

Economic Analysis

LINDSEY MEISCH, JOSHUA LAFOREST, SANAD HWEIJALEXIS
GUILLEN, JOSHUA OLUSOJI

Table of Contents

Capital Cost (OOM).....	1
Equipment Cost – Hand Calculations.....	2
Equipment Cost – Aspen.....	4
ISBL, OSBL, Contingency, Engineering Costs, and Total Capital.....	4
Variable Cost of Production (VCOP).....	6
Fixed Cost of Production (FCOP).....	7
Revenues and Gross Profit.....	8
Simple Pay-Back Period and ROI.....	9
NPV Values at MACRS-5 & MACRS-7.....	9
DCFRROR for MACRS-5 & MACRS-7.....	12
Sensitivity Analysis.....	13
AI Analysis.....	15

Capital Cost (OOM)

First, the capital cost of the methanol production process was estimated using the Order of Magnitude (OOM) approach. This approach relies on purchased equipment costs (PEC) developed from standard cost correlations. The following general cost equation for major equipment was used:

$$C = a + bS^n$$

Where S is the equipment size and constants a , b , and n were obtained from Table 6. 2 in the textbook. When an equipment item was greater than the upper sizing limit of the cost correlation, it was divided into multiple identical units whose individual sizes remained within allowable boundaries. The final plant configuration and associated PECs were calculated accordingly.

Purchase Equipment Cost (PEC)

Eight equipment categories were used in the PEC calculation: heat exchangers, reboiler, compressor, trays, column shell, and two vertical drums. The heat exchangers E-300*, E-301, and E-302 all exceeded the maximum allowable heat-transfer area for the U-tube and thermosyphon correlations. Because of this, they were split into 4-5 parallel shells to keep each unit within the correlation limits.

All PEC calculations used the same sizing basis and were converted to million USD.

Tag	Service / Type	Size	a	n	# shells/units	Per-unit (\$)	Cost per unit (a,S ⁿ) (USD)	PEC total (MM USD)
E-300	Feed-product HX (U-tube S&T)	67	10000	1	1	67	670000	0.67
E-300*	LP steam generator (U-tube S&T)	3020	10000	1	4	755	7550000	30.2
E-301	Condenser (U-tube S&T)	3300	10000	1	4	825	8250000	33
E-302	Reboiler (U-tube S&T)	6130	10000	1	7	876	8760000	61.32
K-300	Recycle compressor (centrifugal)	150	8400	0.6	1	150	169798.5751	0.169798575
T-300 trays	Sieve trays (per tray)	7.9	100	2	23	7.9	143543	0.143543
T-300 shell	Column shell (vertical PV, CS)*	500	230	0.6	1	49500	150831.1198	0.15083112
D-300 & D-301	Drums (vertical PV, CS)*	11900	230	0.6	2	5960	64129.22815	0.064129228

Total Purchased Equipment Cost (PEC)

$$PEC_{total} = 125.72 \text{ MM USD}$$

To convert PEC to installed cost, a Lang Factor of 3.6 was applied:

$$C_{ISBL} = 3.6 \times PEC = 452.59 \text{ MM USD}$$

Additional capital components were estimated as standard percentages of ISBL:

- OSBL = 40% of ISBL
- ECC = 10% of ISBL
- Contingency = 10% of ISBL
- Working Capital (WC) = 15% of ISBL

Cost Component	Basis	Cost (MM USD)
ISBL Installed Cost	$3.6 \times PEC$	452.59
OSBL	$0.40 \times ISBL$	181.03
ECC	$0.10 \times ISBL$	45.26
Contingency	$0.10 \times ISBL$	45.26
Working Capital	$0.15 \times ISBL$	67.89

The Total Capital Cost using the Order of Magnitude Method in this case is the sum of the fixed capital and working capital:

$$TCC = 452.59 + 181.03 + 45.26 + 45.26 + 67.89$$

$$TCC = 792.03 \text{ MM USD}$$

This value utilized the Order of Magnitude method and includes the equipment purchase costs, installation, off-sites, engineering, contingency, and working capital, and represents a $\pm 30\text{-}50\%$ accuracy level typical of early-stage conceptual design.

Equipment Cost – Hand Calculations

Before being able to calculate any of the equipment cost we need to determine what type of equipment should be used for each unit operation. For E-300 and E-300* we use a U shell and tube heat exchanger. E-301 we use a floating head. E-302 we use a kettle as most distillation columns use kettles for their reboilers. T-300 uses a vertical pressure vessel. K-300 uses a reciprocating compressor as these types of compressors are primarily used in recycling units. Then for both D-300 and D-301 also use a vertical pressure vessel

Using the following equation we are able to calculate the equipment costs.

$$C = a + b * S^n$$

The values of a, b, and n are from Table 7.2 in the book Chemical Engineering Design: Principles, Practice and Economics of Plant and Process Design, Second Edition, Elsevier, 2012.

Solving for the pressure values equipment costs, we need to first find the mass of each of the pressure vessels as table 7.2 only helps if we have the mass of the pressure vessel. First need to find the thickness of the pressure vessel wall. Using the pressure values and the diameter size given in the handout we get the following wall thickness.

$$t_w = \frac{P_i * D_i}{4SE - 0.4 P_i} = \frac{0.165 * 7.9}{(4 * 88.96 * 0.7) - (0.4 * 0.165)} = 0.005234$$

With having the wall thickness of T-300 can now solve for the mass of the pressure vessel using the following equation

$$mass = (-t_w^2 + D_i * t_w) * 3.14 * L * \rho = (0.005234^2 + 7.9 * 0.005234) * 3.14 * 12.7 * 7850 = 12953 \text{ Kg}$$

Then reuse the same equation from the beginning to find the total cost of the pressure vessel

Since T-300 is a distillation column we also need to find the cost for each individual tray. Which uses the same equation at the beginning and multiply by 30 since trays can only be bought in sets of 30. Adding them together gives us the total cost of T-300

Repeat the same set of calculations for D-300 and D-301 to find the mass of the pressure vessel and then the equipment cost

Eq-ID	Equipment Name	S	Units	a	b	n	total
E-300	Heat Exchanger (U shell tube and tube)	67	m^2	28,000	54	1.2	\$ 36,388.48
E-300*	Steam Generator (u shell and	3200	m^2	28,000	54	1.2	\$ 896,107.95
E-301	Separator Condensor (floating head shell)	3300	m^2	32,000	70	1.2	\$ 1,199,655.60
E-302	reboiler(kettle)	6130	m^2	29,000	400	0.9	\$ 1,054,119.32
T-300	methanol sepe	7.9	m	130	440	1.8	\$ 666,799.38
K-300	Compressor (reciprocating)	150	kW	260,000	2,700	0.75	\$ 375,726.34
D-300	Liquid-Vapor Separator	18650.269	Kg	11,600	34	0.85	\$ 156,664.53
D-301	Liquid-Vapor Separator	18650.269	kg	11,600	34	0.85	\$ 156,664.53
						TOTAL	\$ 4,542,126.14

Equipment Cost – Aspen

Equipment Costs were calculated using Aspen Process Economic Analyzer V14. The calculations were trivial, simply entering Carbon Steel as the equipment material and their respective data points (like column height, flowrate, heat exchange area, etc.). The following data was found:

Eq-ID	Equipment Name	Model Equipment	Key Data Points	Equipment Costs
E-300	Heat Exchanger (U shell tube and tube)	U-Tube Heat Exchanger	Heat Exchange Area (sq. m): 67	\$28,800.00
E-300*	Steam Generator (U shell and tube)	U-Tube Heat Exchanger	Heat Exchange Area (sq. m): 3200	\$648,500.00
E-301	Seperator Condensor (floating head shell)	U-Tube Heat Exchanger	Heat Exchange Area (sq. m): 3300	\$693,900.00
E-302	Reboiler (kettle)	U-Tube Kettle Type Reboiler	Heat Exchange Area (sq. m): 6130	\$1,498,100.00
T-300	Methanol Seperator	Trayed Tower	Column Diameter (m): 7.9 Column Height (m): 12.7 Stages: 23	\$1,457,800.00
K-300	Compressor (reciprocating)	Reciprocating Compressor	Gas Flow Rate (LPM): 16230	\$987,600.00
D-300	Liquid-Vapor Separator	Vertical Process Vessel	Diameter (m): 2.3 Height (m): 7.0	\$252,600.00
D-301	Liquid-Vapor Separator	Vertical Process Vessel	Diameter (m): 2.3 Height (m): 7.0	\$252,600.00
			TOTAL COSTS	\$5,819,900.00

These costs are reported to 2022's values, which are essentially the same prices for 2025 and thus no cost escalation factor is needed to adjust these values. The total cost calculated is noticeably larger than by hand, which makes sense as the results from hand calculations take cost values from *Chemical Engineering Design (Second Edition)* published in 2013, which has outdated values.

ISBL, OSBL, Contingency, Engineering Costs, and Total Capital

To find the inside the boundary limits we need to take the cost of the equipment and using the CEPCI 2025 to convert the cost into today's currency. The book uses CEPCI 2010 with the cost in the gulf coast. So to convert it use the following equation

$$2025 \text{ Cost} \in \text{Midwest} = \text{Equipment} \frac{\frac{\text{cost} * \text{CEPCI 2025}}{\text{CEPCI 2010}} * 1.2}{1}$$

Take the 2025 price and multiply it by the lang factor for each piece of equipment to get the total cost of equipment with installation. Then add up all the cost of the equipment to get ISBL.

Table 7.4 Installation Factors Proposed by Hand (1958)

Equipment Type	Installation Factor
Compressors	2.5
Distillation columns	4
Fired heaters	2
Heat exchangers	3.5
Instruments	4
Miscellaneous equipment	2.5
Pressure vessels	4
Pumps	4

total	2025 cost in midwest	With Install
\$ 36,388.48	\$ 55,636.07	\$ 194,726.26
\$ 896,107.95	\$ 1,370,101.80	\$ 4,795,356.32
\$ 1,199,655.60	\$ 1,834,210.16	\$ 6,419,735.56
\$ 1,054,119.32	\$ 1,611,692.86	\$ 5,640,924.99
\$ 666,799.38	\$ 1,019,501.09	\$ 4,078,004.35
\$ 375,726.34	\$ 574,465.76	\$ 1,436,164.39
\$ 156,664.53	\$ 239,531.81	\$ 958,127.24
\$ 156,664.53	\$ 239,531.81	\$ 958,127.24
ISBL	\$ 24,481,166.35	

In class Professor Bilgin states that offsite boundary limit (OSBL) is 40% of ISBL, engineering and construction (ECC) is 10% of ISBL, Contingency is 10% of ISBL, working capital (WC) is 15% of ISBL. To find total capital cost use the following equation

$$\text{Total Capital Cost} = \text{ISBL} + \text{OSBL} + \text{Contingency} + \text{ECC} + \text{WC}$$

ISBL	\$ 24,481,166.35
OSBL	\$ 9,792,466.54
Contingency	\$ 2,448,116.64
ECC	\$ 2,448,116.64
Working capital	\$ 3,672,174.95
Total Capital	\$ 42,842,041.12

Variable Cost of Production (VCOP)

To find the variable cost, the amounts of carbon dioxide and hydrogen required for methanol production were determined. From the mass balance provided, 110.6 tons of methanol are produced per hour, while 151.5 tons of carbon dioxide and 21 tons of hydrogen are fed into the system. To determine the consumption ratio, the formula,

Rate of consumption = $\frac{\text{Product}}{\text{Limiting Reactant}}$, was used. In this case, the product is 110.6 tons of methanol, and the limiting reactant is alternated between H₂ and CO₂. The two values were determined to be 1.36 tons of CO₂ and 0.189 tons of H₂ consumed per ton of methanol produced.

From there, the raw material cost was calculated using the USD price per metric ton. Currently, the price of hydrogen in the United States is over \$3,800 per ton; however, if this value were used, the budget would exceed the intended scope of the plant, so alternative options were explored. One option is grey hydrogen, produced from natural gas, which is cheaper than other forms such as green hydrogen. This brings the hydrogen price down to \$2,000 per ton. CO₂ was determined to cost \$48 per ton, and utility costs (steam, electricity, demineralized water, and cooling water) were \$9.60, \$12.50, \$0.22, and \$44.00, respectively. The variable cost was calculated using the formula,

Variable Cost = (H₂ required × H₂ price per ton) + (CO₂ required × CO₂ price per ton) + Utility Cost Sum.

Based on the data, the cost came to \$511.91 per ton of methanol. The annual variable cost was then determined by multiplying the methanol produced per hour (110.6 tons/hr) by the annual operating hours (8,000 hr/yr), giving 884,800 tons per year. This value was multiplied by the variable cost per ton to yield a total variable cost of \$452,855,936 per fiscal year.

CO2 Required	1.369801085 MT	
H2 Required	0.189873418 MT	
Material	Price (USD)	
Hydrogen	2000 MT	
CO2	48 MT	
Steam	9.6 MT	
Electricity	12.5 kWh	
Demineralized Water	0.22 m ³	
Cooling Water	44 m ³	
Variable Cost	511.8172875 USD/ton	
Methanol produced per year	884800 MT/yr	
Variable Cost per year	452855936 USD/yr	

Fixed Cost of Production (FCOP)

For fixed cost calculations, the total capital cost and working capital are used to find the fixed capital investment (FCI). The formula is expressed as $FCI = TCC - WC$. From the previous data, $FCI = 42,842,041.12 - 3,672,174.95 = 39,169,866.16$ US dollars. This value will be used as the basis for calculating equipment costs for each component. Maintenance and repair costs are based on 4% of FCI; operating labor is 1.5%; supervision and overhead are 20% of operating labor; depreciation is 10%; and taxes and insurance are 2.07%. The latter is based on the Illinois property tax rate in 2025. With the exception of supervision and overhead, each component cost is equal to the fixed capital investment multiplied by its associated percentage. For example, maintenance and repairs uses 4% of FCI: $7,234,674.28 \times 0.04 = 156,679.47$ dollars. The calculated values were summed to give a total fixed cost of 7,234,674.28 dollars.

FCI= TCC-WC	39169866.16		
Component	% basis	Cost	
Maintenance & repairs	4% of FCI	1566794.647	
Operating labor	1.5% of FCI	783397.3233	
Supervision & overhead	20% of labor	156679.4647	
Depreciation	10% of FCI	3916986.616	
Taxes & insurance	2.07% of FCI	810816.2296	
Total Fixed Cost		7234674.28	USD

Revenues and Gross Profit

To calculate the revenue, we simply conduct a calculation on the output of methanol production and multiply it by its current selling price. Our methanol plant produces 109.8 tonnes of methanol per hour. Extrapolating this produces:

$$109.8 \frac{\text{tonnes}}{\text{hr}} * 8000 \text{ hrs} = 878,400 \frac{\text{tonnes methanol}}{\text{operating year}}$$

Taking current methanol market values from IMARCGROUP:

$$878,400 \frac{\text{tonnes methanol}}{\text{operating year}} * 1000 \text{ kg} = 878,400,000 \text{ kg} \\ 878,400,000 \text{ kg} * 0.8034 \text{ $ USD} = 705,706,560 \text{ $} = 705 \text{ $ MM}$$

To calculate the gross profit, the CCOP is subtracted from the yearly revenue. The results are:

$$CCOP = FCOP + VCOP = 7.23 \frac{\text{\$ MM}}{\text{yr}} + 194.72 \frac{\text{\$ MM}}{\text{yr}} = 201.95 \frac{\text{\$ MM}}{\text{yr}}$$

$$\text{Gross Profit} = \text{Revenue} - CCOP = 705 \frac{\text{\$ MM}}{\text{yr}} - 201.95 \frac{\text{\$ MM}}{\text{yr}} = 503.5 \text{ \$ MM}$$

Simple Pay-Back Period and ROI

The simple pay-back period is the point at which the full investment is paid for and the net profit is equal to zero. It is calculated as follows:

$$\text{Simple Payback Period} = \frac{\text{Total Investment}}{\text{Average Annual Cash Flow}} = \frac{42.842 \$ MM}{248.94 \frac{\$ MM}{yr}} = .1721 \text{ years after operation} = 2.1721 \text{ years}$$

An additional 2 years were added to the Simple Payback Period to indicate the two years of construction needed for the plant before revenue is generated.

The ROI can be calculated with the following equation:

$$\text{Return on Investment (ROI)} = \frac{\text{Net Annual Profit}}{\text{Total Investment}} \times 100 = \frac{327.94 \$ MM}{42.842 \$ MM} \times 100 = 765.4 \%$$

The return on investment is particularly profitable, as the revenue far exceeds the total investment, or capital costs required to begin operation.

NPV Values at MACRS-5 & MACRS-7

NPV stands for Net Present Value. This is a standard metric used to measure the efficiency/profitability of a project – in this case methanol production. To calculate the NPV, a series of other calculations must first be completed. These include the Pre-Tax Cash Flow, Depreciation, Taxable Income, Cash Flow, and Discount Factor. These calculations were done over a 20-year operational period with 2 years of construction. The calculations using MACRS-5 depreciation are displayed on the following page:

Data

TCC	FCOP	VCOP	Revenue	CCOP	ISBL
42.842	7.235	194.72	705.71	201.955	23.82857
OSBL	ECC	Contingency	WC	Interest Rate	Tax
9.531429	2.382857	2.382857143	3.574285714	0.14	35%

Calculations (MACRS-5)

Years	Pre-Tax Cash Flow(\$MM)	Depreciation	Taxable Income
1	-10.425	0	-10.425
2	-31.275	0	-31.275
3	248.26	7.625142857	240.6348571
4	503.755	12.20022857	491.5547714
5	503.755	7.320137143	496.4348629
6	503.755	4.392082286	499.3629177
7	503.755	4.392082286	499.3629177
8	503.755	2.196041143	501.5589589
9	503.755	0	503.755
10	503.755	0	503.755
11	503.755	0	503.755
12	503.755	0	503.755
13	503.755	0	503.755
14	503.755	0	503.755
15	503.755	0	503.755
16	503.755	0	503.755
17	503.755	0	503.755
18	503.755	0	503.755
19	503.755	0	503.755
20	503.755	0	503.755
21	503.755	0	503.755
22	503.755	0	503.755

Tax	Cash Flow	Discount Factor	Present Value of CF	NPV
0	-10.425	0.877192982	-9.144736842	-9.144736842
0	-31.275	0.769467528	-24.06509695	-33.2098338
0	248.26	0.674971516	167.5684286	134.3585948
84.2222	419.5328	0.592080277	248.3970966	382.7556914
172.0442	331.71083	0.519368664	172.2802107	555.0359021
173.7522	330.002798	0.455586548	150.3448355	705.3807376
174.777	328.9779788	0.399637323	131.4718786	836.8526162
174.777	328.9779788	0.350559055	115.3262093	952.1788255
175.5456	328.2093644	0.307507943	100.9269865	1053.105812
176.3143	327.44075	0.26974381	88.3251153	1141.430927
176.3143	327.44075	0.236617377	77.47817131	1218.909099
176.3143	327.44075	0.207559102	67.96330817	1286.872407
176.3143	327.44075	0.182069388	59.61693699	1346.489344
176.3143	327.44075	0.15970999	52.29555876	1398.784903
176.3143	327.44075	0.140096482	45.87329716	1444.6582
176.3143	327.44075	0.122891651	40.23973435	1484.897934
176.3143	327.44075	0.107799694	35.29801259	1520.195947
176.3143	327.44075	0.094561135	30.96316894	1551.159116
176.3143	327.44075	0.082948364	27.16067451	1578.31979
176.3143	327.44075	0.072761723	23.82515308	1602.144943
176.3143	327.44075	0.063826073	20.89925708	1623.0442
176.3143	327.44075	0.055987783	18.33268165	1641.376882

Calculations (MACRS-7)

Years	Pre-TaxCashFlow(\$MM)	Depreciation	Taxable Income
1	-10.425	0	-10.425
2	-31.275	0	-31.275
3	248.26	5.448164571	242.8118354
4	503.755	9.336987429	494.4180126
5	503.755	6.668187429	497.0868126
6	503.755	4.761901714	498.9930983
7	503.755	3.404626286	500.3503737
8	503.755	3.400813714	500.3541863
9	503.755	3.404626286	500.3503737
10	503.755	1.700406857	502.0545931
11	503.755	0	503.755
12	503.755	0	503.755
13	503.755	0	503.755
14	503.755	0	503.755
15	503.755	0	503.755
16	503.755	0	503.755
17	503.755	0	503.755
18	503.755	0	503.755
19	503.755	0	503.755
20	503.755	0	503.755
21	503.755	0	503.755
22	503.755	0	503.755

Tax	CashFlow	Discount Factor	Present Value of CF	NPV
0	-10.425	0.877192982	-9.144736842	-9.144736842
0	-31.275	0.769467528	-24.06509695	-33.2098338
0	248.26	0.674971516	167.5684286	134.3585948
84.98414	418.7708576	0.592080277	247.9459655	382.3045603
173.0463	330.7086956	0.519368664	171.7597335	554.0642939
173.9804	329.7746156	0.455586548	150.2408786	704.3051725
174.6476	329.1074156	0.399637323	131.5236064	835.8287789
175.1226	328.6323692	0.350559055	115.2050527	951.0338316
175.124	328.6310348	0.307507943	101.0566535	1052.090485
175.1226	328.6323692	0.26974381	88.6465472	1140.737032
175.7191	328.0358924	0.236617377	77.61899235	1218.356025
176.3143	327.44075	0.207559102	67.96330817	1286.319333
176.3143	327.44075	0.182069388	59.61693699	1345.93627
176.3143	327.44075	0.15970999	52.29555876	1398.231829
176.3143	327.44075	0.140096482	45.87329716	1444.105126
176.3143	327.44075	0.122891651	40.23973435	1484.34486
176.3143	327.44075	0.107799694	35.29801259	1519.642873
176.3143	327.44075	0.094561135	30.96316894	1550.606042
176.3143	327.44075	0.082948364	27.16067451	1577.766716
176.3143	327.44075	0.072761723	23.82515308	1601.591869
176.3143	327.44075	0.063826073	20.89925708	1622.491126
176.3143	327.44075	0.055987783	18.33268165	1640.823808

The final values for NPV at year 22 are the following:

$$NPV(MACRS-5): 1641.38 \$ MM$$

$$NPV(MACRS-7): 1640.82 \$ MM$$

As you can see, there is only a small difference in profitability between the two discount rates throughout operation. The reason for this is the high pre-tax cash flow generated from production in comparison to the lower total capital cost. If the revenue was much smaller,

DCFRROR for MACRS-5 & MACRS-7

The discounted cash flow rate of return (DCFRROR) is another metric used to measure the efficiency/profitability of a project, similar to the NPV. Comparing DCFRROR values is generally a better practice, as they consider the number of years needed for construction and for operation. To compute the DCFRROR, the value for NPV at year 22 (the final year of operation) is goalseeked to equal zero by changing the interest rate. The interest rate that results is the DCFRROR. Here are the results for both MACRS-5 and MACRS-7 depreciation rates:

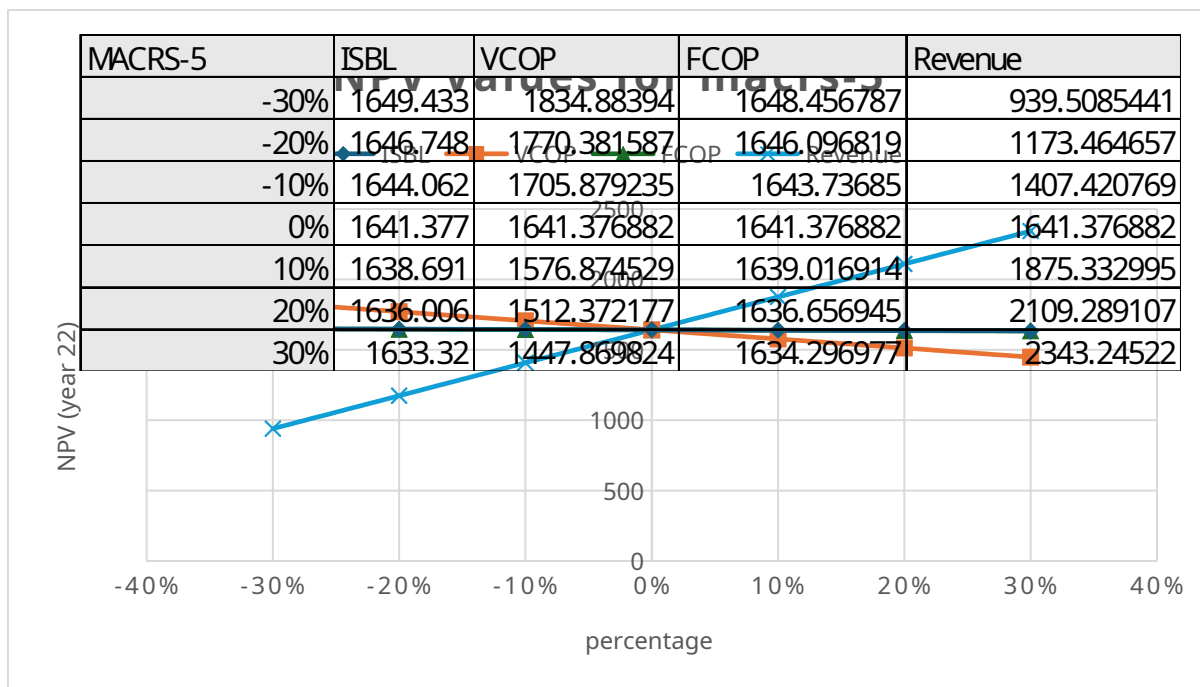
In a similar fashion to the NPV calculations, the DCFRROR values were nearly the same due to the high pre-tax cash flow in comparison to the low total capital cost.

Sensitivity Analysis

A sensitivity analysis is utilized to find how the magnitude of our production changes when different parameters, like the ISBL, revenue, and CCOP are given different values. For our plant, values were varied $\pm 30\%$. The following data was used:

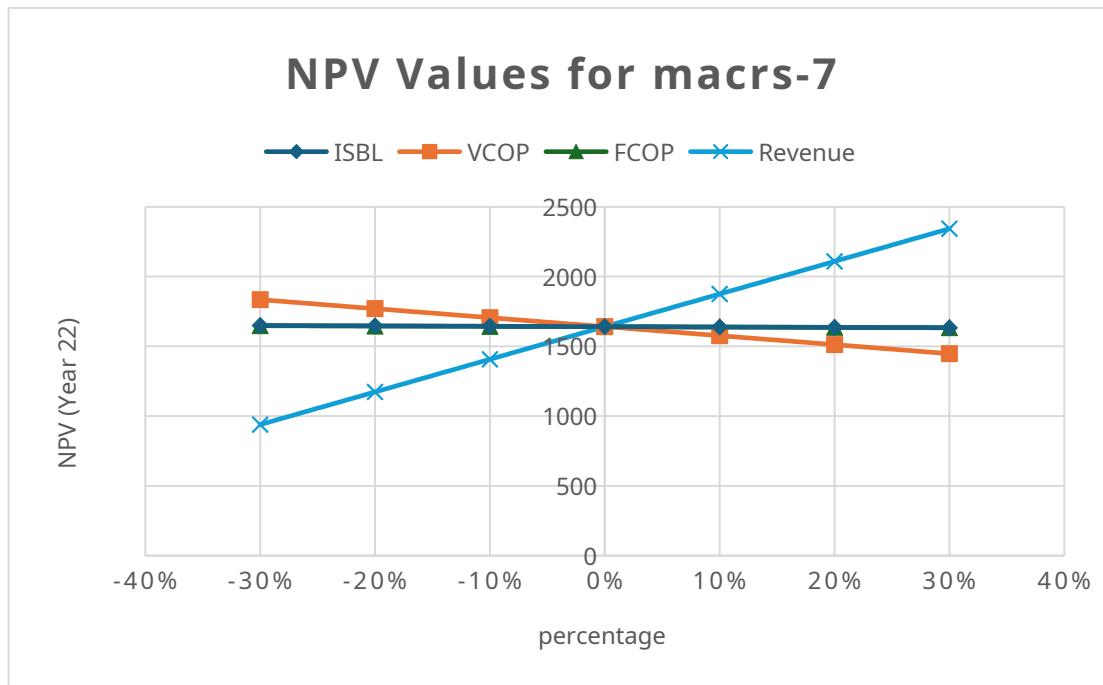
Value	0	-10%	-20%	-30%	10%	20%	30%
ISBL	23.82857	21.44571429	19.06285714	16.68	26.21142857	28.59429	30.97714
OSBL	9.531429	8.578285714	7.625142857	6.672	10.48457143	11.43771	12.39086
ECC	2.382857	2.144571429	1.906285714	1.668	2.621142857	2.859429	3.097714
Contingency	2.382857	2.144571429	1.906285714	1.668	2.621142857	2.859429	3.097714
WC	3.574286	3.216857143	2.859428571	2.502	3.931714286	4.289143	4.646571
TCC	41.7	37.53	33.36	29.19	45.87	50.04	54.21
TCC-WC	38.12571	34.31314286	30.50057143	26.688	41.93828571	45.75086	49.56343
VCOP 50%	97.36	87.624	77.888	68.152	107.096	116.832	126.568
FCOP 50%	7.235	6.5115	5.788	5.0645	7.9585	8.682	9.4055
CCOP 50%	104.595	94.1355	83.676	73.2165	115.0545	125.514	135.9735
VCOP	194.72	175.248	155.776	136.304	214.192	233.664	253.136
FCOP	7.235	6.5115	5.788	5.0645	7.9585	8.682	9.4055
CCOP	201.955	181.7595	161.564	141.3685	222.1505	242.346	262.5415
Revenue 50%	352.855	317.5695	282.284	246.9985	388.1405	423.426	458.7115
Revenue	705.71	635.139	564.568	493.997	776.281	846.852	917.423

These values were imported into the NPV calculations and data on the NPV at year 22 were collected for each parameter varied. The following results were found for MACRS-5:



As expected, decreasing the ISBL, VCOP, and FCOP increase the NPV value, as they directly affect the Pre-tax Cash Flow and Depreciation. Revenue has the exact opposite relationship. From the graph, revenue ends up being the most important factor when it comes to maximizing the NPV, as it has the largest slope. The least important factor to consider is the FCOP, as its original magnitude of 7.235\$ MM is much smaller in comparison to the revenue, and smaller still in comparison to the ISBL and VCOP. A similar trend can be seen for MACRS-7:

MACRS-7	ISBL	VCOP	FCOP	Revenue
-30%	1648.88	1834.330866	1647.903713	938.9554701
-20%	1646.195	1769.828513	1645.543745	1172.911583
-10%	1643.509	1705.326161	1643.183776	1406.867695
0%	1640.824	1640.823808	1640.823808	1640.823808
10%	1638.138	1576.321455	1638.46384	1874.779921
20%	1635.453	1511.819103	1636.103871	2108.736033
30%	1632.767	1447.31675	1633.743903	2342.692146



MACRS-7 depreciation offers almost identical results to MACRS-5, although the slopes are slightly smaller for each varied factor. Again, this can be explained from the fact that depreciation is dependent on TCC (total capital costs), and the TCC in this case is much smaller in magnitude in comparison to the cash flow. Thus, depreciation does not change our NPV value as heavily as it would if our process was less profitable (lower cash flow).

AI Analysis

Tool: Gemini 2.5 Pro

Prompt: I need to analyze the economic feasibility of a new chemical manufacturing plant that produces methanol via carbon hydrogenation. I have attached the process flow diagram, material balance table, major equipment details and utilities table as well as table 6.2 to get the capital cost using order of magnitude. For the heat exchangers, break them up into identical smaller sizes and then sum them up. Compute the following items: a) Capital cost with order of magnitude method. b) Equipment costs. c) ISBL, OSBL, contingency, engineering costs. d) Total capital cost. e) Variable cost of production (raw materials and utilities) forecasted prices for raw materials and products. f) Fixed cost of production (assume that a year equals 8000 hours. This is about 330 days, which allows for periodic shut-down and maintenance). g) Revenues and gross profit. h) Simple pay-back period and ROI. i) NPV value calculated at a discount rate of 14%, using MACRS depreciation with a 7 year recovery term, and 5-year recovery term. j) Discounted cash flow rate of return using MACRS depreciation with a 7-year recovery term, and 5-year recovery term. k) Sensitivity Analysis on capital cost, utilities, raw material cost and product cost. Assume construction will take 2 years, and the capital will be paid in two equal increments, at the end of the first and second years. The savings (income) can be taken as paid at the end of each year. Production will be 50% in first start on the completion of construction.

Output: I can begin the preliminary equipment cost calculations, but I have encountered significant missing data and data incompatibilities that prevent me from completing the full economic analysis.

Below is the partial calculation for the equipment costs (Task b) for the items where data was available and compatible, followed by a detailed list of the information I still need.



Partial Equipment Cost Calculation (Task b)

The cost for each piece of equipment is calculated using the formula: $\text{Cost} = a + bS^n$, based on the parameters in Tables 6.2.

Note on Heat Exchangers: As you instructed, for exchangers where the required size (S_{total}) exceeded the table's upper limit (S_{Upper}), I have split the duty into N identical smaller units. N is the smallest integer such that $(S_{\text{total}}/N) \leq S_{\text{Upper}}$.

Equipment	Description	Sizing Value (S)	Data Source (Ta	a	b	n	Calculation Details	Estimated Cost
E-300	Feed-Product HEX	67 m ²	Exchanger, U-tube	10,000	115	1	$10,000 + 115 \cdot (67)^{1.0}$	\$17,705
E-300*	LP Steam Generator	3020 m ²	Exchanger, U-tube ($S_{\text{Upper}}=1000$)	10,000	115	1	N=4 units @ 755 m ² each. $4 \cdot [10,000 + 115 \cdot (755)^{1.0}]$	\$387,300
E-301	Condenser	3300 m ²	Exchanger, U-tube ($S_{\text{Upper}}=1000$)	10,000	115	1	N=4 units @ 825 m ² each. $4 \cdot [10,000 + 115 \cdot (825)^{1.0}]$	\$419,500
E-302	Reboiler	6130 m ²	Thermosyphon reboiler ($S_{\text{Upper}}=500$)	13,000	95	1	N=13 units @ 471.5 m ² each. $13 \cdot [13,000 + 95 \cdot (471.5)^{1.0}]$	\$751,352
K-300	Recycle Compressor	150 kW	Compressor, Centrifugal	8,400	3,100	0.6	$8,400 + 3,100 \cdot (150)^{0.6}$	\$66,959



Missing & Incompatible Data

I cannot calculate the cost for the reactor, column, or separators, which are critical for the Total Capital Cost. Here are the specific issues:

R-300 (Reactor): The equipment details table does not provide a sizing parameter (e.g., volume in m³) for the R-300 reactor. The cost data in Table 6.2 requires volume to estimate the cost.

T-300 (Vessel) & D-300/D-301 (Vessels): The equipment list provides dimensions (height and diameter in meters). However, the cost data in Table 6.2 for "Pressure vessels" requires the shell mass in kg as the sizing parameter S. I cannot calculate the shell mass from dimensions alone.

T-300 (Trays): The column diameter is listed as 7.9 m. The cost data for all tray types (sieve, valve, bubble cap) has a maximum valid size (S_{Upper}) of 5.0 m. The equipment is too large for the provided cost model.



Information Required for Full Analysis

To complete your request (Tasks c through k), I need the following information.

1. Capital Cost Data (Tasks c, d)

Missing Equipment Sizes: Please provide the volume (m³) of R-300 and the shell mass (kg) for T-300, D-300, and D-301. (Or, provide alternative cost data compatible with dimensions).

Costing Factors:

Cost Index: What is the base year for the costs in Table 6.2, and what is the target year for the analysis (e.g., CEPCI for 2010 and 2024)?

ISBL/OSBL Factors: Factors for piping, instrumentation, civil, etc. (ISBL) and utilities, storage, etc. (OSBL), usually as a percentage of the Total Purchased Equipment Cost (PEC).

Indirect Cost Factors: Percentages for Engineering and Contingency (e.g., % of ISBL+OSBL).

2. Operating Cost Data (Tasks e, f)

Raw Material Prices:

Price of "Captured CO₂" (in /tonne)

Price of "H₂ Feed" (in /tonne)

Utility Prices:

Price of Cooling Water (in /MT)

Price of LP Steam (in /MT)

Price of Electricity (in /kWh)

Fixed Cost Factors:

Operating Labor: Number of operators per shift and the loaded labor rate (/hr).

Maintenance: Typically as a percentage of Total Capital Cost (TCC).

Overhead, Taxes, Insurance: Typically as a percentage of TCC or labor.

3. Revenue Data (Task g)

Product Price:

Price of "Grade AA Methanol" (in /tonne)

4. Financial Parameters (Tasks h, i, j)

Project Lifespan: The number of years to run the discounted cash flow analysis (e.g., 15 or 20 years).

Corporate Tax Rate: The tax rate (%) to be applied to profits.

Working Capital: The amount of capital required to be held, or a method to estimate it (e.g., 15% of annual revenue).

Once you can provide this missing information, I can proceed with the complete techno-economic analysis.

Prompt: For missing inputs, estimate it logically using standard chemical engineering design heuristics and Table 6.2 correlations (assume U-tube shell-and-tube exchangers, centrifugal compressor, carbon-steel pressure vessels, and sieve trays). Clearly state each assumption. Use a Lang Factor of 3.6, OSBL is 40% of ISBL, ECC is 10% of ISBL, Contingency is 10% of ISBL, and Working Capital is 15% of ISBL. Use a 21% tax rate and 14% discount rate. If any dimension is not provided, estimate it from diameter/height using a 20 mm wall thickness and steel density = 7850 kg/m^3 .

Output: This is a comprehensive techno-economic analysis based on your provided data and the requested standard chemical engineering design heuristics and factors.

I have broken down the analysis into the required parts, starting with the assumptions and the equipment/capital costs.

Assumptions and Heuristics

To complete the analysis, the following standard chemical engineering design heuristics and factors were used:

ID	Description	Value/Source
A1	Reactor Volume (R-300)	500 m3 (assumed maximum size for cost correlation)
A2	T-300 Column Split	Split into 2 columns of 4.0 m diameter each (since 7.9 m exceeds SUpper of 5.0 m for trays)
A3	Vessel Sizing	Calculated mass using wall thickness $t=0.02$ m and steel density $\rho=7850$ kg/m3
A4	Heat Exchanger Splitting	E-300*, E-301, E-302 were split into multiple identical units to meet the SUpper limit of Table 6.2
A5	Operating Hours	8,000 hr/year
A6	Estimated Prices	CO2:\$30/tonne; H2:\$2,000/tonne; CW:\$0.05/MT; LP Steam:\$10/MT; Electricity: \$0.10/kWh; Methanol:\$400/tonne
A7	Labor & Fixed Costs	4 operators/shift; Labor Rate:\$100,000/person/year; Maintenance: 5% of TFCI; Operating Supplies: 10% of Maintenance; Laboratory: 10% of OLC; Overhead: 60% of OLC; PTI:2% of TFCI

a) - d) Capital Cost Summary

b) Equipment Costs

The cost for each piece of equipment is calculated using the order-of-magnitude method: $\text{Cost}=a+b \cdot S^n$. Equipment costs are summarized in the table below.

Equipment	Type	Units	S (Actual)	S Unit	a	b	n	Cost(USD)
E-300	HEX (U-tube)	1	67	m2	10,000	115	1	17,705
E-300*	HEX (U-tube)	4	755	m2	10,000	115	1	387,300
E-301	HEX (U-tube)	4	825	m2	10,000	115	1	419,500
E-302	Reboiler (Thermosyphon)	13	471.5	m2	13,000	95	1	751,352
K-300	Compressor (Centrifugal)	1	150	kW	8,400	3,100	0.6	66,959
R-300	Reactor (Jacketed)	1	500	m3	14,000	15,400	0.7	1,732,311
T-300 (Vessel)	Vessel (Vertical, CS)	1	49,444.00	kg	-400	230	0.6	758,256
T-300 (Trays)	Sieve Trays	46	4	m (dia.)	100	120	2	92,920
D-300/301	Vessel (Vertical, CS)	2	7,977.00	kg	-400	230	0.6	228,095
Total Purchased Equipment Cost (PEC)								4,452,398

a), c), d) Total Capital Cost

The total capital investment (TCI) is calculated using the provided Lang Factor and multiplier rules.

Item	Calculation Basis	Cost (USD)
Total Purchased Equipment Cost (PEC)	Sum of Equipment Costs	\$4,452,398
ISBL Installed Cost (CISBL)	PEC×3.6 (Lang Factor)	\$16,028,633
OSBL	40%×CISBL	6,411,453
Engineering & Construction (ECC)	10%×CISBL	1,602,863
Contingency	10%×CISBL	1,602,863
Total Fixed Capital Investment (TFCI)	CISBL+OSBL+ECC+Contingency	\$25,645,813
Working Capital (WC)	15%×CISBL	2,404,295
TOTAL CAPITAL INVESTMENT (TCI)	TFCI+WC	\$28,050,108

e) Variable Cost of Production

The plant produces 494,400 tonne/year of CH₃OH based on \$61.8 tonne/hr x 8000 hr/year.

Raw Materials

Component	Flow (tonne/hr)	Flow (tonne/year)	Price (/tonne)	Cost (/year)
CO ₂ (201)	151.5	1,212,000	\$30	36,360,000
H ₂ (202)	21	168,000	\$2,000	336,000,000
Total Raw Material Cost (TRMC)				\$372,360,000

Utilities

Utility	Usage	Unit	Price (/unit)	Cost (/year)
Cooling Water	7330 MT/hr×8000 hr	MT	\$0.05	2,932,000
LP Steam	150 MT/hr×8000 hr	MT	\$10	12,000,000
Electricity	150 kW×8000 hr×1 kW/h	kWh	\$0.10	120,000
Total Utility Cost (TUC)				\$15,052,000

TOTAL VARIABLE COST OF PRODUCTION (CV)		\$387,412,000
--	--	---------------

f) Fixed Cost of Production

Cost Item	Calculation Basis (A7)	Cost (/year)
Operating Labor Cost (OLC)	12 people×\$100,000/person	1,200,000
Maintenance	5%×TFCI	1,282,291
Operating Supplies	10%×Maintenance	128,229
Laboratory	10%×OLC	120,000
Plant Overhead	60%×OLC	720,000
Property Taxes & Insurance (PTI)	2%×TFCI	512,916
TOTAL FIXED COST OF PRODUCTION (CF)		\$3,963,436

g) Revenues and Gross Profit

Item	Calculation Basis	Value (/year)
Annual Revenue (R)	494,400 tonne×\$400/tonne	\$197,760,000
Total Cost of Production (CT)	Variable Cost+Fixed Cost	387,412,000+3,963,436=\$391,375,436
Gross Profit	R-CT	197,760,000-391,375,436=\$-193,615,436

The project yields a negative gross profit due to the high estimated cost of H₂ raw material compared to the revenue from Methanol, indicating the operation is fundamentally economically unfeasible under these price assumptions. The subsequent financial analysis will still be performed for completeness, but it will confirm the high probability of failure.

h) Simple Pay-back Period and ROI

Simple Return on Investment (ROI)

$$\text{ROI} = \frac{\text{Annual Net Profit}}{\text{Total Capital Investment (TCI)}} \times 100\%$$
 Since H_2 cost dominates, we'll use $\text{Net Profit} = \text{Gross Profit}$ to show the impact.

$$\text{ROI} = \frac{-\$193,615,436}{\$28,050,108} \times 100\% = -689\%$$
 ### Simple Pay-back Period

$$\text{Pay-back Period} = \frac{\text{TCI}}{\text{Annual Cash Flow}}$$

$$\text{Annual Cash Flow} = \text{Gross Profit} + \text{Depreciation}$$
 (assuming zero tax, which is the case with negative profit). Since the annual cash flow is negative, the project **never pays back** the initial investment.

---- ## i) & j) Discounted Cash Flow Analysis (DCF) The analysis will use a 14% discount rate and a 21% tax rate. Construction takes 2 years, with 50% of TFCI paid at the end of Year 1 and the remaining 50% at the end of Year 2. Production starts in Year 3 at 50% capacity, and Year 4 onwards at 100% capacity.

Item	Year 0	Year 1	Year 2	Year 3	Year 4+
TFCI Cash Out	\$12,822,907	-\$12,822,907			
WC Cash Out/In			-\$2,404,295	\$2,404,295	
Revenue (\$R)				\$98,880,000	\$197,760,000
Operating Cost (\$C _O)			\$195,687,718	\$391,375,436	
Profit Before Tax (PBT)			-\$96,807,718	-\$193,615,436	

The project's PBT is negative, so the Tax is $\$0$ (no taxes are paid, and no tax shield from negative profit is considered for this simplified analysis).

7-Year MACRS Depreciation The MACRS depreciation percentages for a 7-year class life are: 14.29% , 24.49% , 17.49% , 12.49% , 8.93% , 8.92% , 8.93% , 4.46% (8 years).

Year	T (MACRS)	Depreciation	Cash Flow	NPV (14%)
0				
1	14.29%	\$3,664,575	-\$12,822,907	-\$11,248,164
2	24.49%	\$6,280,517	-\$12,822,907	-\$9,864,688
3	17.49%	\$4,485,346	-\$92,322,372	-\$62,374,258
4	12.49%	\$3,203,771	-\$190,411,665	-\$112,658,573
5	8.93%	\$2,289,957	-\$191,325,479	-\$99,576,648
6				
7				

Since the annual cash flow becomes highly negative starting in Year 3, the **NPV is highly negative** ($<-\$400$ Million) over 12 years, and the **Discounted Cash Flow Rate of Return (DCFROR) is $<0\%$** (meaning it's not even equal to the cost of capital).

5-Year MACRS Depreciation The MACRS depreciation percentages for a 5-year class life are: 20.00% , 32.00% , 19.20% , 11.52% , 11.52% , 5.76% (6 years). The results are qualitatively identical: $\text{NPV} \ll 0$ and $\text{DCFROR} \ll 14\%$.

---- ## k) Sensitivity Analysis The project's negative profitability is overwhelmingly due to a single input: the high cost of H_2 compared to the revenue from $CH_3\text{OH}$.

Breakeven Analysis The H_2 cost component of the $\text{Total Variable Cost}$ is $\$336$ Million. The Total Revenue is $\$197.76$ Million. For the project to break even ($\text{Gross Profit} > 0$), the Total Cost

of \$H₂ would need to be reduced by \$194 Million\$ (the amount of the negative gross profit) or the \$Methanol\$ price increased by a similar amount. \$

$$\text{Required Breakeven } H_2 \text{ Price} = \frac{\text{Total Revenue} - \text{Other Costs}}{\text{Annual } H_2 \text{ Flow}}$$

$$\text{Other Costs} = \text{TRMC}(\text{CO}_2) + \text{TUC} + \text{TCF} = \$36.36 \text{M} + \$15.052 \text{M} + \$3.963 \text{M} = \$55.375 \text{ Million}.$$

$$\text{Required Breakeven } H_2 \text{ Price} = \frac{\$197,760,000 - \$55,375,436}{168,000 \text{ tonne/year}} \approx \$847/\text{tonne}$$

Variable	Change for Breakeven	Sensitivity
H2 Cost (\$2000/tonne)	Must drop to \$847/tonne (a 57.6% drop)	HIGH
Methanol Price (\$400/tonne)	Must increase to \$790/tonne (a 97.5% increase)	HIGH
Capital Cost (TFCI)	TFCI has minimal impact on Gross Profit	LOW

Conclusion:

The project is highly sensitive to the H₂ cost and Methanol price. At the assumed H₂ price of \$2,000/tonne, the plant is not economically viable. The H₂ cost would need to be nearly halved for the operation to break even on an annual basis before considering taxes and depreciation.

Reflection

Gemini actually performed pretty well in structure, completeness, and clarity. It was able to produce a full techno-economic workflow with the assumptions through sensitivity analysis and didn't require much prompting other than requesting that it make reasonable assumptions. It was able to apply chemical-engineering heuristics like the Lang factor method, heat exchanger splitting, 8000 hr/yr operation, and MACRS depreciation. It used reasonable default prices for raw materials and utilities and generated detailed numerical tables. With some slight formatting issues, it was able to organize the sections and made the workflow easy to follow. The output showed clear process economics logic and its ability to propagate consistent assumptions and

perform a full cash flow with depreciation schedules, demonstrated that it has a strong analytical capability and familiarity with engineering-economic conventions.

That said, Gemini struggled with data contextualization and units. Several of the inputs were assumed, like reactor volume and CO₂ and H₂ prices without distinguishing which values came from the provided tables vs which were estimates. The formulas used were correct, but it seemed to have mis-applied the constants a, b, and n from Table 6.2. It treated them as additive instead of multiplicative and occasionally mixed \$/MT, \$/tonne, and \$/1000 lb\$ units. These discrepancies show a limitation in engineering judgement. Gemini can apply formulas but not always verify whether the magnitude of its results is realistic. It also didn't check the economic plausibility until the end, where it reported a huge negative profit. Human engineers would have noticed earlier that the H₂ price dominated the economics.

Another observed weakness was the inflexibility with incomplete data. Gemini stalled or 'invented' placeholder values when key parameters like reactor volume and vessel mass were missing instead of estimating them from geometry immediately. While it produced a full NPS and DCFROR analysis, the calculations became meaningless once profitability turned negative. Gemini still generated discount cash flow tables without interpreting that the project should be abandoned at the gross profit stage.

In conclusion, Gemini excelled at following procedure and producing a coherent, reproducible economic analysis with professional formatting and logical flow. Its biggest limitations were engineering realism and contextual awareness. It followed equations flawlessly but lacked the intuitive sense to validate assumptions or flag unrealistic cost magnitudes early. It can be deduced that when AI tools such as Gemini are used as a computational assistant, it can dramatically accelerate the report preparation, but it still requires human oversight to ensure technical and economic soundness.